

Summery

Computer engineering PhD (Dec 2025) specializing in low-level kernel/system/architecture optimization for LLM training/inference across GPUs and custom accelerators. Built cycle-accurate perf models, LLM inference compute/communication kernels on GPU/NPU/custom accelerators. Expertise in efficient AI deployment, including parallelism, quantization, sparsity, attention optimizations, and computer architecture, including compute engine, memory hierarchy and scale-up/out network.

Education

University of Michigan

Ph.D., Computer Science and Engineering Department

Ann Arbor, MI, USA

Sep. 2020 - Dec. 2025

Zhejiang University (ZJU)

B.Eng., College of Information Science and Electronic Engineering

Hangzhou, China

Sep. 2016 - June 2020

Professional Experience

University of Michigan

Graduate Research Assistant, Advisor: Reetuparna Das

Ann Arbor, MI, USA

Sep. 2020 - Dec. 2025

Co-design architecture and system for large-scale emerging applications, including Generative AI and genome sequencing. Publish papers at top conferences including ISCA, ASPLOS and HPCA, awarded as Communication of ACM Research Highlights (24/10,000+) and HPCA Distinguished Artifact Honorable Mention (3/29).

- Profile throughput, power and GPU utilization (Nsight Compute) for LLM inference using vLLM on DGX-H100.
- Investigate system optimizations for LLM deployment, including parallelism (DP/TP/PP/EP/SP/CP), model sharding (ZeRO/FSDP), quantization, sparsity, speculative decoding and attention optimization (GQA/MLA/Flash-Attn).
- Design a Compute Express Link (CXL)-enabled Processing-In-Memory (PIM) system for LLM inference, generating $5.2\times$ tokens/\$ than A100 GPUs. Open source tools for operation-level workload mapping and cycle-accurate simulations (Obtain >80 ★ on GitHub). *This system is under patent filing and technique transfer to SK Hynix's next generation data center.* Design a data-reuse aware PIM architecture, generating $8.9\times$ tokens/\$ than H100 GPUs (*Under patent filing*). Design a task-level scheduler for long context LLM inference on heterogeneous GPU/NPU/PIM cluster. (*Ongoing*)
- Design a low-power GPU architecture using dynamic voltage scaling (DVS) and tape-out a prototype. Propose a checkpoint-based fault-tolerant system to mitigate the error caused by DVS. (*Ongoing*)
- Create a genomics benchmark suite, characterizing microarchitectural and memory access patterns, as well as instruction diversity and thread scaling. Design a domain specific accelerator for dynamic programming algorithms. This accelerator is evaluated by a cycle-accurate simulator and achieves $157.8\times$ throughput/mm² than A100 GPUs.

Tenstorrent Inc.

Performance Architect Intern, Manager: Wei-han Lien

Santa Clara, CA, USA

May 2023 - Aug. 2023

Build a profile-guided agile performance model for Tensix AI accelerator, including computation and communication between RISC-V cores, with $<10\%$ error rate. This model reduces the performance modeling time from a few hours to ~ 5 minutes, enabling fast kernel development and optimization.

Intel Labs

Graduate Research Intern, Manager: Nilesch Jain

Hillsboro, OR, USA

June 2022 - Aug. 2022

Develop an offline profile and online prediction-based resource scheduler for Quality of Service (QoS)-aware job co-location for ML inference, achieving up to $1.4\times$ request throughput improvement over single model deployment and reducing scheduling times from ~ 30 minutes to ~ 10 seconds compared to state-of-the-art schedulers.

Publications & Recognitions (Selected)

CACM Research Highlight for GenDP (ISCA 2023) — among 24 selected across 10K+ papers.

HPCA Distinguished Artifact Honorable Mention (2025)

Outstanding Young Speaker Award (APPT 2025)

Top Conference Publications - ISCA*/ASPLOS*/HPCA/ISPASS/IISWC (* First-author)

Technical Skills

Programming: C/C++, Python, CUDA, Verilog, Bash

Tools: Linux, Docker, Git, Slurm, VTune/Pin, Nsight Compute/System, NCCL, VCS, Synopsis Design Compiler

Framework: PyTorch, vLLM

Under-Review Publications

- [1] Data Reuse-Aware Processing-In-Memory Architecture for Efficient Large Language Model Inference. (*Under Submission to ASPLOS'26, first author*)
- [2] Fast Resource Management for Quality of Service-Aware Co-Location of Machine Learning Inference. (*Under Submission to SIGMETRICS'26, first author*)
- [3] Addressing In-Memory Database Bottlenecks with CXL Memory Expansion. (*Under Submission to ASPLOS'26*)

Peer-Reviewed Publications

* equal contribution

- [1] Noah Kaplan, Jan-Niklas Schmelzle, **Yufeng Gu**, Christopher Batten, Reetuparna Das. "PangenomicsBench: A Benchmark Suite and Characterization of Pangenomics." IISWC'25.
- [2] **Yufeng Gu**, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." (Invited Paper) Communications of the ACM, 2025.
- [3] Alireza Khadem*, Kamalavasan Kamalakkannan*, Zhenyan Zhu, Akash Poptani, **Yufeng Gu**, Jered Benjamin Dominguez-Trujillo, Nishil Talati, Daichi Fujiki, Scott Mahlke, Galen Shipman, Reetuparna Das. "DX100: Programmable Data Access Accelerator for Indirection." (ISCA'25).
- [4] **Yufeng Gu***, Alireza Khadem*, Sumanth Umesh, Ning Liang, Xavier Servot, Onur Mutlu, Ravishankar Iyer, Reetuparna Das. "PIM Is All You Need: A CXL-Enabled GPU-Free System for Large Language Model Inference." (ASPLOS'25). **In Progress of Transferring to SK Hynix's Next Generation Datacenter**
- [5] Alireza Khadem, Daichi Fujiki, Hilbert Chen, **Yufeng Gu**, Nishil Talati, Scott Mahlke, Reetuparna Das. "Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing." (HPCA'25). **Distinguished Artifact Honorable Mention (3/29)**
- [6] **Yufeng Gu**, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." (ISCA'23). **Communication of ACM Research Highlights (24/10,000+)**
- [7] Arun Subramaniyan, **Yufeng Gu**, Timothy Dunn, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, and Reetuparna Das. "GenomicsBench: A Benchmark Suite for Genomics." (ISPASS'21).
- [8] Xiaoxiao Li, **Yufeng Gu**, Nicha Dvornek, Lawrence H. Staib, Pamela Ventola, and James S. Duncan. "Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results." Medical Image Analysis 65: 101765. *IF = 11.1*
- [9] Xiaoxiao Li, Yuan Zhou, Nicha C. Dvornek, **Yufeng Gu**, Pamela Ventola, and James S. Duncan. "Efficient Shapley Explanation for Features Importance Estimation Under Uncertainty." (MICCAI'20).